

Team Name	Nationalities	Institutions	Mentor	Problem description
2Bcities	Peru/Iran	Univ Geneva, PARIS21	Shashansk Agarwal	We are platform to rethink, question and explore what it takes to build Resilient Future Cities. We want our cities 2B inclusive, 2B sustainable and 2B smart. As for a challenge we are addressing the following: How can cities build digital skills, trust, and confidence so residents (especially marginalized groups: migrant women) can effectively use digital public services?
5factorial!	Bangladesh	Bangladesh University of Professionals	Viviana Umpierrez	We are currently addressing the following connectivity challenges:1. Limited network coverage in rural districts2. Affordability gaps3. Subpopulations excluded due to low digital skills.
Amarjargal Batzorig	Mongolia	Smart Software Group, Mobile Network Operator / Telecommunications Company (Mongolia)	Amarnath Ojha	Answer: Our project addresses the challenge of uneven and inefficient network coverage in geographically extreme and sparsely populated regions of Asia, where harsh climate, desert terrain, mountains, and long travel distances limit the economic viability of traditional network expansion. We focus on identifying under connected and underserved user clusters by analysing network performance, service quality, and user behavior patterns. Our goal is to optimize the placement of network infrastructure (base stations and coverage sites) to maximize population coverage, improve service reliability, and reduce investment inefficiencies. By combining user level demand analysis with geographic and network constraints, we aim to support data-driven decisions for equitable and cost effective connectivity expansion.
Bengal Nomads	Bangladesh	Daffodil International University, Shahjalal University of Science and Technology, Chittagong University of Engineering and Technology (CUET)	Amarnath Ojha	Despite rapid growth, a stark urban-rural divide leaves millions in remote Bangladesh offline due to high infrastructure costs and limited network coverage. Marginalized rural youth face a secondary usage gap where high data prices and lack of personal devices make essential digital learning largely unaffordable. Low digital literacy further prevents many communities from using existing connections for e-agriculture or healthcare. Our project identifies these specific connectivity barriers to design data-driven policies that bridge the gap for a truly inclusive and meaningful digital future.
Bridge the Bit	Bangladesh	Shahjalal University of Science and Technology	Nathan Menton	With a particular emphasis on the Gender-Based Usage Gap, our project tackles Bangladesh's "Second-Level Digital Divide." Although 99% of people have access to 4G and 5G infrastructure, national data from 2025–2026 shows a notable "Usage Desert": more than 50% of people are not online, and there is a record gender disparity in the use of mobile internet. We are tackling the latent barriers that prevent women and marginalized groups from having meaningful connectivity even though they have physical access to a network because of: 1. Device Affordability: Smartphones have a higher cost-to-income ratio than feature phones. 2. Digital Literacy Elasticity: The inability to independently navigate services other than USSD/Voice. In order to reveal the socioeconomic "deserts" concealed within covered areas, we seek to identify geographical clusters where high signal strength negatively correlates with actual data consumption.
BridgingBytes	Bangladesh	Morgan Stanley, ICICI Lombard	Esperanza Magpantay	We are addressing the challenge of lack of meaningful connectivity in areas that are officially classified as connected but continue to experience poor service quality, low broadband speeds, and high affordability barriers. This issue is particularly evident in rural districts, peri-urban regions, and low-income urban settlements, where vulnerable subpopulations remain digitally excluded due to a combination of inadequate infrastructure performance, cost constraints, and limited digital skills, leading to persistent and often invisible digital inequality. By combining data on network performance, service usage patterns, demographic vulnerability, and socio-economic indicators, we aim to identify communities where connectivity exists in theory but does not translate into real digital participation. This shifts the policy lens from infrastructure deployment alone to evidence-based targeting of subsidies, digital skills programs, public Wi-Fi investments, and inclusive service design, ensuring connectivity leads to actual social and economic inclusion rather than nominal coverage.
BudakByte	Malaysia	Malaysian Communications and Multimedia Commission	Amarnath Ojha	Addressing gaps in meaningful connectivity caused by inconsistent broadband service quality, which limits users' ability to reliably access essential digital services such as video streaming, web browsing, online voice and video calls, and online gaming.
BUP_Kamikaze	Bangladesh	Bangladesh University of Professionals	Viviana Umpierrez	"My analysis of the 2023 ITU dataset reveals a significant Digital Divide in Bangladesh characterized by structural inequalities across three primary dimensions: Geography, Generation, and Gender. Geographic Disparity: A delta of 16.95% exists between Urban (35.47%) and Rural (18.52%) participation rates, indicating persistent infrastructure and accessibility bottlenecks in non-metropolitan regions. Generational Exclusion: The data highlights a severe generational gap of 32.91%; while youth (15-24) engagement is robust at 33.93%, senior citizens (75+) are statistically marginalized at a near-zero adoption rate of 1.02%, suggesting a critical need for targeted digital literacy intervention. Gender Variance: A baseline gap of 5.32% between male and female users remains, necessitating gender-responsive connectivity policies. Bf Image will not show the you can visit this linky utilizing these high-fidelity insights, I aim to develop a predictive framework for identifying 'connectivity deserts' to optimize resource allocation and ensure inclusive digital growth."
Connect 404	Bangladesh	Shahjalal University of Science and Technology	Shashansk Agarwal	We are tackling the crisis of "Digital Deserts" in topographically challenging regions—specifically the total network blackouts found in remote tea gardens and hill tracts. While many urban areas debate internet speeds, these regions face a complete lack of both mobile and data coverage. This challenge creates two critical problems: 1. Data Invisibility: Because there is no signal, these populations leave no digital footprint, making them invisible to development planners and policymakers. 2. Economic and Safety Risks: During our own fieldwork in the tea gardens of Sylhet, our team experienced the safety risks of being unable to communicate in a "dead zone." For the local ethnic communities, this isolation prevents them from accessing digital markets, mobile banking, and essential services, trapping them in a cycle of economic exclusion despite their proximity to major cities. Our project aims to bridge this digital divide by identifying these infrastructure blind spots and proposing ways to bring "Universal and Meaningful Connectivity" to populations that the current grid has physically left behind.
Connectiva	Bangladesh	Rajshahi University of Engineering & Technology, MBM Goup, Bangladesh Telecommunication Regulatory Commission	Akanksha Sharma	We are addressing the following issues associated with UMC: - Digital Skills and Literacy: How to Use the Internet, What to learn, How and Where to Learn from, How to be Safe Online.- Devices: Appropriate Device Ownership for Meaningful Use.- Affordability: internet access, devices and data plansWe plan to tackle these issues in the rural parts of Bangladesh.

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ConneXions	Spain/Morocco	Ecole Polytechnique, Paris	Amarnath Ojha	Our project addresses the "Intensity Gap", a critical dimension of the Usage Gap highlighted in the ITU Facts and Figures 2025 and the GSMA State of Mobile Internet Connectivity 2025 reports. While infrastructure (coverage) now reaches 96% of the global population, the real "digital desert" exists where people are technically "covered" but cannot meaningfully participate in the digital economy. We are specifically targeting the Asia-Pacific SIDS (Small Island Developing States) context, where the challenge is three-fold: The Quality Disparity: According to latest 2025 estimates, a typical user in a high-income country generates nearly eight times more mobile data than one in a low-income country. This "intensity gap" proves that current connectivity in SIDS is often insufficient for anything beyond basic messaging, excluding populations from high-value activities like telehealth or remote work. The Affordability Barrier: We are addressing the "SIDS Penalty," where broadband costs remain 46% higher than the global median, rendering even 5G-covered areas "digital deserts" for the bottom 40% of the population. The Resilience Gap: As noted in the 2025 Baku Action Plan, connectivity in the Pacific is highly vulnerable to climate shocks. We are focusing on identifying areas where a lack of redundant infrastructure creates "fragile connectivity," preventing a safe and resilient digital environment.
CyberCat's	Ukraine	State University of Communication and Informational Technologies	JL Cervera-Ferri	Optimizing Satellite Connectivity Allocation during Urban Blackouts. We address the critical loss of mobile communication in densely populated areas of Kyiv during scheduled power outages. When grid power fails, terrestrial base stations get overloaded or shut down. Specific Challenge: Satellite providers (like Starlink) or emergency responders lack real-time, granular geospatial data to know exactly where to direct capacity or deploy ground terminals. Our solution identifies "connectivity voids" - hyper-local zones where high population density intersects with active power outages-providing exact coordinates to target satellite coverage efficiently.
D4A(Data for all)	Kenya	St Paul University	Shashansk Agarwal	Our team, D4A, addresses the challenge of limited and unreliable network coverage in Nyakach Sub-county. Many areas experience weak signals and frequent disruptions, making meaningful Internet use difficult. This coverage gap especially affects students, farmers, women, and rural households, limiting access to education, digital services, and economic opportunities. Our focus is on identifying these connectivity blind spots and translating real community experiences into evidence for targeted network expansion and inclusive digital policies.
Data^3 (Data cubed)	Armenia	Armenian State University of Economics	Amarnath Ojha	A real-life example from our community illustrates a key digital divide issue. Recently, an elderly man went to the passport office and was told that he needed to register through an online system. However, he could not do it because he does not even have a touchscreen phone. The staff suggested taking a photo and completing it at home with help, but he didn't have a camera either. Sudden digitization of public services without proper guidance or consideration for vulnerable populations, such as the elderly, leads to exclusion and prevents full access to essential services. This is a real example that occurred in our community, showing that some subpopulations are excluded due to lack of digital skills or access to devices, even when digital services are theoretically available. Similar challenges are faced in many countries around the world, making this a widespread, global digital inclusion problem.
Data5	Armenia	French University in Armenia (UFAR)	Nathan Menton	Affordability-based digital exclusion: many users have internet access but cannot afford or pay for premium educational and productivity tools, leaving them connected yet unable to fully participate in digital learning and knowledge work.
DataLemurs	Canada/poland	British Columbia Institute of Technology (BCIT), KartGis, Breeze Energies Sp. z o.o.	Esperanza Magpantay	We will tackle the reasons for area-dependent lack of access to the internet.
DataTron	Bangladesh	Shahjalal University of Science and Technology	Fredrik Eriksson	We would like to address the following problems: Affordability gap, Sub-population excluded due to low digital skills.
Digital Bridge Team	State of Palestine	Palestine Technical University - Kadoorie - Faculty of Information Technology and Artificial Intelligence	JL Cervera-Ferri	We are addressing a compound connectivity challenge in Palestinian refugee camps, which combines three main problems exacerbated by unique political and social conditions: First, a severe infrastructure deficit due to political restrictions and dense urban geography, creating real "digital deserts" where network coverage is very limited or non-existent. Second, an affordability crisis, where the cost of connectivity represents a significant portion of the income of households suffering from high unemployment and poverty, making the internet inaccessible even when it is available. Third, the exclusion of specific community groups such as women and girls (due to cultural barriers), youth (deprived of opportunities for distance education and work), and the elderly (due to the digital skills gap). In short, the problem is not just a weak signal or high cost alone, but an intertwined crisis of lacking infrastructure, poverty, skills, and political restrictions combined.
Digital bridges	Malawi	Spatial Girls Network, Malawi Revenue Authority, Old Mutual Malawi	Esperanza Magpantay	Unstable and limited connectivity in remote areas affordability gap slack of technical expertise.
EduBandwidth	Bangladesh	University of Frontier Technology, Bangladesh	JL Cervera-Ferri	The project focuses on identifying digital deserts affecting students from low-income and rural communities, where inadequate ICT infrastructure, slow internet speeds, and high data costs prevent meaningful access to online learning platforms and digital educational services.
Fortidata	Cambodia	ABA Bank, United World Schools (UWS)	JL Cervera-Ferri	Internet access in Cambodia has grown rapidly. By 2023, about 79.4% of the population was online, showing strong progress in connectivity and infrastructure reported by ITU. But access does not mean understanding. Many people use the internet every day, yet lack the skills to use it safely and effectively. UNESCO data shows that in 2020, only about 32% of Cambodians had basic digital skills, while very few had intermediate or advanced abilities. Skills like checking reliable information, protecting personal data, or using digital tools for work and learning are still limited. This gap has real consequences. Misinformation spreads easily, online scams are increasing, personal data is at a high risk, and many people miss the opportunities in job seeking. Therefore, the gap between internet usage and digital literacy should be addressed as soon as possible.

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GenDev	Bhutan	GovTech Agency	Amarnath Ojha	We plan to focus on the government e-service accessibility gap in remote areas caused by poor network infrastructure and low digital literacy among rural populations. Our team works in different sections of a government agency that provides online services to citizens, and through our daily work, we have consistently observed a critical challenge: despite successfully digitizing numerous government services, there remains a significant gap between the services we make available online and the citizens who can actually access them. This gap is particularly acute in remote and rural areas where two interconnected barriers prevent meaningful connectivity: (1) inadequate network infrastructure with limited or unstable mobile broadband coverage that makes accessing online government services unreliable or impossible, and (2) insufficient digital literacy where many rural citizens lack the digital skills needed to navigate and utilize government online platforms effectively. This dual barrier means that essential government services—from healthcare registration and social welfare benefits to business permits and educational resources remain inaccessible to those who often need them most, creating a situation where digitization, intended to improve service delivery, inadvertently deepens inequality. We, therefore, aim to systematically map these accessibility gaps, identify the most underserved populations and locations, and develop evidence-based policy recommendations that address both infrastructure deficits and digital skills gaps to ensure truly universal access to government services.
GIS4TELCOM	Mozambique	Mozambique Nacional Communication Institute - INCM	Nehal Desai	Identify areas in Mozambique with the greatest gaps in Universal Meaningful Connectivity (UMC), focusing on the intersection of limited network coverage, connectivity quality, affordability, and digital skills.
GJU Team	Jordan	German Jordanian university	Shashansk Agarwal	The project focuses on the problem of (digital deserts), where people are unable to benefit from the Internet in a meaningful way. In some areas, connectivity is limited because network coverage is weak or missing, while in other areas the infrastructure exists but people still cannot use the Internet properly due to high costs, lack of devices, or low digital skills. Instead of looking at connectivity as a single issue, the project aims to understand the different reasons why people are digitally excluded. By considering both coverage gaps and usage-related barriers, we try to identify the real challenges that prevent communities from accessing and using ICT services effectively.
Guardians of the sea	Brazil	Universidade Estadual de Campinas	Akanksha Sharma	Our project addresses the critical connectivity divide in the remote offshore islands of Bangladesh, specifically focusing on isolated regions like Char Kashem. These areas face severe limited network coverage, inconsistent signal strength, and a lack of reliable broadband infrastructure due to their geographical isolation and vulnerability to natural disasters. By analyzing current network types and reliability, we aim to identify technical and affordability gaps that exclude these populations from essential digital services. Our goal is to propose solutions that ensure stable, high-speed connectivity for these underserved coastal communities.
Hedgehog	Cambodia	Cambodia Academy of Digital Technology, Royal5G	Akanksha Sharma	The project addresses the connectivity and digital exclusion gaps faced by marginalized subpopulations in rural, low-income provinces. Specifically, we are targeting affordability gaps and exclusion due to limited digital skills among high-risk youth, the elderly, and persons with disabilities, which hinders their access to education and safe employment.
HerRise	Bangladesh	Bangladesh Maritime University, Bangladesh University of Business and Technology (BUBT)	Nehal Desai	Coastal area females are deprived of mobile banking services Low levels of formal education among women Lack of digital literacy Comparatively low income levels Lower rate of mobile phone ownership among women Prevailing patriarchal social norms High priced internet Poor and limited network coverage in coastal districts Low broadband speeds in the coastal area Affordability gaps, Subpopulation excluded due to low digital skills
iamgroot / SUST_iamgro	Bangladesh	Shahjalal University of Science and Technology	Nehal Desai	We are addressing the affordability gap (Income Gap) in Bangladesh, which is a critical dimension of Universal and Meaningful Connectivity (UMC) we will focus on the income gap, which prevents large segments of the population from accessing digital services. Service Affordability & Device Availability: Many people cannot afford smartphones, internet subscriptions, or data plans, which limits their ability to participate in the digital economy, access educational resources, or benefit from e-government and health services. Closing this affordability gap is essential for achieving Universal and Meaningful Connectivity (UMC), ensuring that connectivity is not just available, but truly usable and impactful for everyone, regardless of income level.
IME Brasil	Brazil	INSTITUTO MILITAR DE ENGENHARIA	Viviana Umpierrez	Online risks for Brazilian children due to lack of knowledge.
Inlix	Uganda	Nkumba University	Viviana Umpierrez	We are addressing the urban–rural connectivity “usage gap” and infrastructure mismatch in Uganda. While major urban centers like Kampala have good 4G coverage, many rural areas remain “digital deserts” facing two main challenges: 1. Coverage Gap: Some communities have no network access at all or rely on outdated 2G networks. 2. Usage Gap: In areas where coverage exists, high poverty levels, low smartphone ownership, and limited digital skills prevent people from using the internet meaningfully. Using census and local data, our project identifies these underserved communities and prioritizes them for combined infrastructure investment and affordability support, ensuring connectivity is both available and usable.
Life Connect	Bangladesh	Bangladesh Maritime University	Fredrik Eriksson	Dublar Char, a remote island in the Sundarbans, faces significant internet deprivation due to extremely weak, unreliable, or non-existent signals. The backbone of income in Dublar Char is fishing and fish processing, almost entirely dependent on the Bay of Bengal and the Sundarbans ecosystem. Our project addresses all around connectivity exclusion in Dublar Char, where physical isolation, extreme climate exposure, urban-rural market price difference and socioeconomic vulnerability intersect. Mobile signals are weak, unstable, and often absent due to lack of nearby towers, backhaul infrastructure and power reliability. Even when signal exists, speeds are insufficient for meaningful use such as video calls, telemedicine, online education or real-time weather alerts. Existing data packages are priced for urban markets, making them unaffordable for seasonal, low-income fishing households. Fishermen at sea, women involved in dried fish or 'shutki' processing, children and elderly residents are excluded due to: Lack of digital literacy, language barriers, no locally relevant digital services. Absence of real-time internet restricts access to: Marine weather forecasts, cyclone early-warning systems, emergency coordination and disaster response. Especially in safety sectors, fishermen can only rely on radio before leaving for work or mouth-heard unreliable weather forecasts. In an essence, the challenge is not just sustainable internet access but meaningful, safe, affordable and resilient connectivity in a climate-vulnerable island economy.

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Linka	Ghana/Niger/Kenya	BigData Ghana Limited, Gill Technologies	Amarnath Ojha	Linka is addressing affordability-driven digital deserts in rural and peri-urban Kenya, where mobile broadband coverage technically exists but remains meaningfully inaccessible due to low incomes, high data costs, limited electricity access, and low digital skills. While coverage maps suggest these areas are connected, actual internet use remains low. This creates a hidden digital divide where populations are counted as connected but are effectively offline. Our focus is on identifying "invisible digital deserts" that are masked by national connectivity averages and coverage statistics.
Merczcord Technologies	Nigeria	MERCZCORD TECHNOLOGIES	Nathan Menton	Our project addresses the hidden digital divide in rapidly urbanizing and peri-urban areas, where network coverage exists but connectivity is not meaningful. Specifically, we focus on "Digital Deserts", geographical areas and subpopulations that are officially classified as connected, yet remain digitally excluded due to a combination of poor service quality, limited affordability, unreliable infrastructure, low digital skills, and socioeconomic constraints. We target three interlinked challenges: Low-quality connectivity in high-density urban and informal settlements, where broadband speeds, latency, and network reliability are insufficient for education, work, or digital public services. Affordability gaps, where the cost of devices, data bundles, and electricity relative to household income prevents regular and effective internet use, even where coverage is available. Exclusion of vulnerable subpopulations, particularly low-income households, women, youth, and residents of informal or rapidly expanding settlements, who are left behind due to limited digital skills and unequal access to ICT services. By combining geospatial, connectivity, and socioeconomic data, our project moves beyond coverage maps to identify, classify, and explain why meaningful connectivity fails, enabling more targeted and equitable digital inclusion policies.
MOHONA	Bangladesh	ACI Logistics Limited	JL Cervera-Ferri	We are addressing the lack of ICT access in rural and hard-to-reach areas of Bangladesh. Our primary focus is poverty as an initial factor that may limit access to broadband and mobile data. After collecting data from the ICT Ministry, we aim to identify additional variables that also contribute to digital exclusion.
Najah Link	Israel/Palestine	An-Najah National University	JL Cervera-Ferri	Our project focuses on underserved communities in Arab and MENA regions, where limited network coverage, affordability issues, and gaps in digital skills prevent meaningful internet access. Addressing these challenges is essential to reduce the digital divide and enable equal access to education, employment, and digital services.
NeuronData	Cambodia	Amret, Manulife Cambodia	Shashansk Agarwal	The connectivity challenge we are addressing is limited and unreliable internet access for students in rural education areas, compounded by unequal access to digital devices at the household level. Specifically: Many rural villages, communes, and districts have poor or no network coverage, making internet access available only within school compounds or not at all. Where connectivity exists, slow and unstable internet speeds disrupt online and blended learning, reducing student engagement and learning effectiveness. Students often cannot access educational content outside school hours, limiting their ability to revise lessons, practice ICT skills, or stay informed. Device scarcity within households (multiple family members sharing one or two phones) further excludes students from consistent digital participation, even when some connectivity is present. As a result, students in these areas experience digital exclusion from e-learning and ICT education, creating education digital deserts and widening the learning gap between urban and rural populations.
NextPlease	State of Palestine	Birzeit University	JL Cervera-Ferri	For this purpose we want to identify "digital deserts" in the West Bank in refugee communities (camps and surrounding neighborhoods) in which people may have nominal coverage yet they are not substantially connected due to a combination of the following: Affordability (concerns of cost of data + income needs). Quality problems (Slowness of service, Congestion, Unstable Service). Digital skills barriers (especially for youth and women in some contexts). Safety/security limitations (online harms, harassments, privacy, etc.) Problem statement: Our plan will include mapping and describing the broadest location of the usage gap in the West Bank, particularly focus on refugee regions, and approximating which kinds of barriers (affordability, quality, skills, safety) have the most significant effects on propelling underusage despite coverage.
No One Offline	India/Uzbekistan/India	Global Law Forum, Tashkent University of Information Technologies, club management about startup in India	Akanksha Sharma	Our idea addresses the challenge of "invisible digital deserts" - contexts where internet coverage and device access formally exist, yet connectivity is not meaningful. Despite high coverage indicators, many individuals and communities are unable to translate access into real social and economic outcomes due to low digital skills, lack of trust and safety, limited perceived relevance, affordability barriers, and quality constraints. As a result, official connectivity metrics often overestimate actual digital inclusion and mask persistent forms of exclusion. Another problem is the lack of enshrined digital human rights at the international and national levels, so that everyone has high-quality, meaningful, and universal access to the internet (so that people consciously understand that internet access correlates with their personal well-being, this will help solve the growing problem of trust in technology).
Nompum	Cambodia	Cambodian University of Technology (CamTech)	Esperanza Magpantay	We aim to address the limited network coverage in North-eastern Cambodia, which significantly affects indigenous communities in Ratanakiri, Mondulkiri, Kratie, Stung Treng, Kampong Thom, and Preah Vihear. According to a report by CamboJA News, rural and indigenous communities still struggle with internet access. Samorn Pich, a 23-year-old Bunong woman from Mondulakiri, noted that her community faces a dual challenge: poor connectivity and a lack of digital literacy. Consequently, when NGOs provide training or workshop opportunities, that information often fails to reach them. Furthermore, Pin Plii, 22, also from the Bunong community, emphasizes the internet's role as a vital business tool. He believes that with reliable access and digital training, indigenous people could sell produce online and bypass traditional, often less profitable, sales methods. Beyond economics, this lack of coverage creates a safety crisis. Tourists and residents in these remote areas are left vulnerable during emergencies, unable to contact authorities or the police. Critically, without stable internet, these communities miss early warning alerts for floods or disease outbreaks, leaving the people of the North-east dangerously uninformed.
NoSQL No Cry	Bangladesh	Bangladesh University of Professionals	Fredrik Eriksson	The project addresses the challenge of hidden digital exclusion within urban areas that are officially classified as well-connected. While network coverage and signal availability are high, significant segments of the urban population remain excluded from meaningful connectivity due to affordability constraints, low-capability devices, and limited digital skills. This challenge manifests as "invisible digital deserts"-communities that appear connected in infrastructure statistics but are unable to effectively use the internet for essential services such as online education, e-government platforms, digital health, and economic opportunities. The gap is not driven by lack of network access, but by poor usability and unequal digital readiness, which prevents real participation in the digital society.

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OK Nepal / Open Knowledge	Nepal	Open Knowledge Nepal	Nathan Menton	We are addressing Nepal's terrain-induced coverage paradox: areas that are officially listed as "connected" to mobile networks but experience little or no usable connectivity due to mountainous geography. Despite statistics claiming over 109 percent of the total population mobile coverage, millions of people in hill districts remain functionally disconnected. The challenge arises from three main factors. First, terrain shadow zones, where settlements are within the theoretical range of towers (5–10 km) but ridges and valleys block signals. Second, signal degradation zones, where weak or unstable 2G/3G connections are insufficient for essential services like online education, telemedicine, digital payments, or e-governance. Third, the quality–affordability trap, where poor 4G signal quality forces repeated data recharges, making connectivity effectively unaffordable despite reasonable base prices. Together, these factors leave millions of Nepalis "invisibly disconnected", counted as connected in official statistics but excluded from meaningful digital participation.
ORBITz	Cambodia	SBI Group	Akanksha Sharma	We are addressing the connectivity disparity between urban and rural areas in Cambodia, where urban residents have reliable broadband access while approximately 75% of the population living in rural areas faces limited connectivity. Specifically, we will examine: - Quality disparities in internet speed, reliability, and consistency between urban centers and rural provinces - Infrastructure gaps in remote and indigenous areas (Monduliri, Ratanakiri regions) - The gap between technical coverage and meaningful connectivity due to affordability and digital literacy barriers - The "last mile" problem where communities are technically covered but lack meaningful connectivity due to quality, affordability, or digital literacy barriers, moving beyond simple coverage metrics to measure true digital inclusion.
PLASTICONN	Nigeria	PlastiConn, Univ of Lagos	Akanksha Sharma	Team PLASTICONN is addressing affordability gaps in relation to connectivity costs that are transient in quality and riddled with low broadband speeds across different areas in Lagos. The average Nigerian youth has story for days about how the cost of 'data' is enough to purchase an abode or go on vacation. Despite the high cost of connectivity, young people that rely heavily on fast broadband speed for their means of livelihood have to deal with slow broadband speed that put their jobs and even education at risk, as well as transient quality of data with no concrete explanation from telecommunication service providers. Our project focuses will on understanding and mapping these connectivity inequalities by analyzing internet costs, broadband quality, and coverage across different areas in Lagos Nigeria. We also aim to identify areas with poor or no internet access the "digital deserts" and compare major Internet Service Providers (ISPs) to reveal where affordability and quality gaps are most severe.
QRARG-AiBot	Bangladesh	Daffodil International University	Viviana Umpierrez	We are responding to the existing gap in accessible, affordable, and high-quality internet connectivity for underserved communities, especially in rural and peri-urban areas where investment in infrastructure has been lower and quality has been inconsistent. The challenge specifically includes: Inadequate last-mile infrastructure in rural and geographically distant areas, leading to unreliable or no broadband connectivity. Low bandwidth and network congestion in densely populated, impoverished urban areas, where shared infrastructure is insufficient to handle demand. Affordability gaps, where the cost of devices and monthly data packages eats disproportionately into household budgets. Digital exclusion of vulnerable groups, including women, youth, senior citizens, and informal sector workers, who lack both connectivity and digital literacy skills to effectively engage online. Unreliable service delivery and robustness, especially in areas with power instability or environmental disruptions. This connectivity gap directly impacts the use of education, healthcare, financial services, employment, and digital governance—locking in socioeconomic disparities. Our response is not only about increasing physical network reach but also about providing affordable, robust, and community-driven connectivity solutions that enable meaningful and sustained digital inclusion, not just basic or temporary connectivity.
RAT	India	The Maharaja Sayajirao University of Baroda	Shashank Agarwal	Low broadband speeds in urban slums, urban inequality gaps, internet accessibility in low income areas, digital literacy across multiple populations, factors influencing income inequality, scope for improvements
Respawn	Bangladesh	United International University (UIU)	Fredrik Eriksson	We are addressing the multi-dimensional digital divide in Bangladesh, where despite very high mobile network coverage, meaningful connectivity remains limited due to several factors. These include geographic barriers (hill and riverine/delta areas with difficult terrain and seasonal flooding), speed disparities between rural and urban areas, affordability gaps for low-income households, and infrastructure inefficiencies such as uneven tower distribution creating digital deserts.
Rocket League	India	BHARATI VIDYAPEETH COLLEGE OF ENGINEERING, Birla Institute of Technology - Mesra, Mesra	Amarnath Ojha	We are addressing the gap between available telecommunication infrastructure and people's ability to meaningfully use it. This includes limited or low-quality network coverage in rural and remote areas, affordability barriers, low digital skills, and social exclusion of vulnerable groups. The challenge lies in the lack of AI-driven, people-centred analysis that connects telecom data with real human needs, leading to underutilized networks and persistent digital exclusion.
SawaNet	Kenya	Jomo Kenyatta University of Agriculture and Technology (JKUAT)	Esperanza Magpantay	Our project addresses the digital divide driven by both limited network coverage and affordability constraints in Kenya. We focus on: Limited network coverage in underserved rural counties, where insufficient infrastructure restricts access to reliable mobile and broadband connectivity. Affordability gaps affect low-income populations, particularly urban low-income youth, where connectivity exists but remains underutilized due to high costs of devices and data plans. By combining these dimensions, the project highlights how connectivity is not only about availability, but also about the ability to meaningfully access and use digital services.
Seeping lights	Bangladesh	Independent University, North South University, Delivery Engineering, Dhaka University	Nehal Desai	We are going to focus on the specific connectivity status of the rural remote areas called the tea states like Sylhet, Chittagong, and Rangpur (in Bangladesh), where the digital divide exists. Based on the data, we will identify and analyze the causes of the digital divide and bottlenecks.
Someting 1.0	Bangladesh	BRAC University, Bangladesh Maritime University	Nathan Menton	I am addressing the challenge of climate-induced digital exclusion in coastal Bangladesh, where weak, unreliable, and disrupted connectivity prevents communities from communicating urgent needs during cyclones, floods, and water crises. In areas like Patakhali, poor network reliability limits access to early warnings, government services, and humanitarian support, leaving local suffering underreported and increasing vulnerability during emergencies.
Statonomics	Bangladesh	Statonomics	Nehal Desai	1. Improper Network infrastructure 2. Less Knowledge in rural Areas 3. Not having up to the mark consumer devices 4. More Demand of Bandwidth than Supply 5. Improper Network planning in urban and rural places

Team Name	Nationalities	Institutions	Mentor	Problem description
Team Alristocrats	Nepal	Kathmandu University	Akanksha Sharma	We are addressing the "Meaningful Connectivity Reliability Crisis" in Nepal. While official NTA (Nepal Telecommunications Authority) MIS reports suggest a broadband penetration rate of over 140%, this is a "statistical mirage" caused by multi-SIM ownership. Real-world data (NLSS 2022/23 and 2025 updates) shows that actual household internet adoption is closer to 40-50%. Specifically, we are focusing on the Fixed-Mobile Divide: rural users rely on unstable, high-latency 2G/3G/4G mobile data, which prevents "meaningful" use (like high-definition remote education or telehealth), whereas urban centers have access to 80+ Mbps FTTH (Fiber-to-the-Home). Our project aims to identify "Digital Deserts" where physical 4G coverage exists, but quality and affordability remain below the ITU's meaningful connectivity thresholds.
Team QBits	India	Jaypee Institute of Information Technology	Esperanza Magpantay	The Invisible Crisis: Quality-Based Digital Exclusion in Nominally "Connected" Areas Our proposal addresses a critical blind spot in Universal and Meaningful Connectivity (UMC) measurement: hidden digital deserts that exist within areas reporting adequate network coverage. While traditional connectivity metrics successfully identify macro-level infrastructure gaps—villages without cell towers, regions without fiber—they systematically fail to detect micro-geographic zones where nominal connectivity fails to deliver meaningful access. This creates a measurement paradox: communities appear "digitally included" in national statistics and coverage maps while residents experience chronic connectivity poverty that prevents participation in online education, telemedicine, digital employment, and civic engagement. These populations remain invisible in policy discourse because existing indicators aggregate away the very heterogeneity that defines their exclusion. Research Focus: Three Dimensions of Hidden Digital Deserts1. Indoor Coverage Deserts: The Building-Level Digital Divide The Phenomenon National coverage maps indicate network availability based on outdoor signal propagation models and tower proximity. However, actual usability depends on indoor penetration, which varies dramatically based on: Building materials: Concrete walls (40-60 dB attenuation), metal roofs (50-70 dB attenuation), reinforced structures Building density: Urban canyon effects, interference from neighboring structures Spatial heterogeneity: Floor level (ground vs. upper floors vs. basements), room position, window orientation Micro-topography: Local terrain features, vegetation density, proximity to interference sources Why This Matters for UMC Education: Students in different classrooms of the same school experience vastly different connectivity—ground floor rooms may have usable service while upper floors or basements become digital exclusion zones, creating intra-school inequality Healthcare: Rural clinics with thick concrete walls or metal roofing cannot conduct telemedicine consultations despite being within tower coverage radius, forcing patients to travel hours for basic consultations Residential: Multi-story buildings show extreme floor-to-floor variation; some residents cannot access online government services, remote work, or distance learning that neighbors use routinely Economic: Small businesses in certain buildings cannot leverage e-commerce, digital payments, or online marketing while competitors 100 meters away thrive digitally The Statistical Gap Current datasets measure outdoor coverage or use sparse indoor sampling that misses building-level and room-level heterogeneity. This creates false confidence in universal access while millions remain functionally disconnected in "covered" areas. Our Analytical Approach Data sources to integrate: Cell tower locations + signal propagation models Building footprints from satellite imagery + OpenStreetMap User-generated speed test data (geolocated, timestamped) Crowd-sourced connectivity quality reports School/clinic/hospital location databases Demographic data on building occupancy patterns Methods: Spatial interpolation of user-generated speed tests to estimate indoor vs. outdoor performance gaps Machine learning classification of buildings by connectivity quality based on construction type, height, distance to towers Geospatial clustering to identify "connectivity inequality streets" where adjacent buildings show >10x speed differences Statistical modeling of signal attenuation by building characteristics using regression analysis Key metrics to develop: Indoor-outdoor connectivity quality ratio by building type Percentage of educational/healthcare institutions with inadequate indoor connectivity despite nominal coverage Building-level connectivity Gini coefficient to measure intra-neighborhood inequality Critical infrastructure connectivity score (schools, hospitals, government offices) 2. Temporal Deserts: Peak-Hour Congestion and Service Degradation The Phenomenon Network capacity is shared among users, creating temporal patterns where nominal bandwidth becomes unusable during critical periods: Morning peak (7-10 AM): School hours coincide with maximum congestion, reducing available bandwidth by 60-80% precisely when students need it for online classes and assessments Evening peak (6-9 PM): After-work hours see bandwidth collapse, making homework completion, skill development courses, remote work, and family communications impossible Weekend congestion: Saturdays/Sundays show different usage patterns, often better for education/work activities Disaster spikes: During emergencies, network overload prevents critical communication when it's most needed for coordination and safety Seasonal patterns: Monsoon/festival seasons create predictable congestion or service degradation Why This Matters for UMC Educational inequality: Students who can only access the internet during peak hours (after school, after parents' work) experience degraded service that renders interactive learning platforms, video lectures, and virtual labs unusable—they appear "connected" but cannot use digital education Work-from-home barriers: Remote
Team SHUSH	Bangladesh	United International University	Nehal Desai	Our project focuses on the usage gap in Bangladesh, where mobile broadband coverage exists but meaningful connectivity is limited due to poor service quality and affordability constraints. We aim to identify district-level digital deserts where internet speeds and reliability are insufficient for education, work, and access to digital public services
TECHDEV-ZIMBABWE	Zimbabwe	TECHDEV-ZMBABWE, RAMANI, Minerva Risk	JL Cervera-Ferri	We are addressing digital deserts caused by uneven network coverage, poor quality of connectivity, and affordability barriers, particularly in rural, peri-urban, and low-income communities across Africa. Specifically, the challenge includes: Limited or absent mobile and broadband network coverage in rural and remote areas
TeleBengal Analyst	Bangladesh	Bangladesh University of Professionals	Nehal Desai	We address the gender digital divide in rural Bangladesh, where adult women face low internet adoption despite mobile network availability. The challenge is driven by affordability of smartphones and data, limited digital skills, and lack of meaningful use cases linked to income, health, and public services. As a result, connectivity exists but remains under-used and not meaningful, limiting women's social and economic participation.
TERA	Jordan	University of Jordan , Mutah University	Shashansk Agarwal	The project addresses the challenge of hidden and future digital exclusion by identifying, monitoring, and predicting "digital deserts" areas where connectivity is insufficient to enable universal and meaningful connectivity despite nominal network availability In the first phase, the project constructs a Digital Desert Index "DDI" that captures the multidimensional nature of connectivity by combining indicators of network coverage, broadband speeds (fixed and mobile), affordability, internet usage, and economic context. This allows the identification of countries and regions where digital deprivation exists today but may be overlooked by traditional access based metrics. In the second phase, the project applies predictive modeling and time-series analysis to forecast the evolution of the "DDI" under multiple scenarios (best-case, baseline and worst case) this enables the development of an early warning system that identifies countries and populations at risk of future deterioration in meaningful connectivity. In the final phase : the project integrates a policy layer that translates analytical insights into actionable policy recommendations, supporting targeted interventions such as infrastructure investment affordability measures and digital skills development by linking measurement prediction and policy guidance the project enables decision makers to move from reactive monitoring to proactive prevention of digital exclusion.

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The Bayesian Scouts	Bangladesh	Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh	JL Cervera-Ferri	Identifying digital deserts with multidimensional gaps in meaningful Internet use in Bangladesh.coverage exists but quality is poor due to weak backhaul, congestion, terrain and building interference, and high data prices relative to incomeUsage and behavioral gaps disproportionately affecting women, elderly populations, and low-literacy groups despite nominal access.Here is a National Research that shows that the Gap: Bangladesh-National-ICT-Household-Survey.pdfAnother Recent (2025) research indicates to the strong existence of the problem: Bridging the Last Mile: Unpacking the Rural Digital Divide in Bangladesh
The Delta Analysts	Bangladesh	University of Frontier Technology, Bangladesh	Nathan Menton	Our team is addressing the critical infrastructure loss and connectivity displacement caused by severe river erosion in the delta regions of Bangladesh.Unlike general rural coverage issues, our challenge is unique because it involves:Physical Infrastructure Destruction : When riverbanks collapse, they take down mobile towers and underground fiber cables and leading to a sudden and also total digital blackout for entire communities.Connectivity Displacement : Displaced populations who migrate to new chars or temporary shelters find themselves in dark zones with no network coverage, hindering their access to emergency aid and early warning systems.Economic Fragility : The survivors of erosion lose their livelihoods, making even basic internet services completely unaffordable, thus widening the digital divide. We are using Data Science and Geospatial Mapping to pinpoint these vulnerable disconnected zones and propose resilient solutions for Universal Meaningful Connectivity (UMC) .
Signal Seekers	Zimbabwe	Institute of Statistical Research and Training	Fredrik Eriksson	Low Broadband Speeds in Urban slums and Affordability Gaps
The signal seekers	Jordan	Applied Science Private University	Fredrik Eriksson	the challenge we're going to tackle is going to be limited network coverage in urban districts and low-speed networks or no networks at all at some times in urban places
The Silent Bandwidth	Bangladesh	Bangladesh University of Professionals	JL Cervera-Ferri	We are tackling digital exclusion based on identity in Bangladeshi Rohingya refugee camps.There is mobile network coverage, and legal SIM cards are now available to refugees. However, the lack of a portable digital identity or simple KYC system prevents them from making meaningful use of the internet.Humanitarian organizations have biometrically registered them, but this identity is incompatible with other online services.Because of this, refugees cannot easily access important digital services such as: Education platforms Digital health information Mobile money or assistance programs Safe online communicationCoverage and SIMs are available, but identity barriers prevent actual digital participation, resulting in a significant usage gap.
University of Buenos Aires	Argentina/Spain/Italy	Universidad Nacional de La Plata, Aerolíneas Argentinas, MS Estudio Contable	Viviana Umpierrez	We address the paradox of urban digital deserts within the Buenos Aires Metropolitan Area (AMBA), home to 14 million people and world-class digital infrastructure. Despite official connectivity statistics showing over 90% internet penetration, 3.2 million residents in peri-urban settlements (low income settlements) remain digitally excluded. These populations rely on expensive prepaid mobile plans – costing three times more per megabit than fixed broadband – whilst receiving speeds below 10 Mbps, far beneath Universal and Meaningful Connectivity thresholds. This isn't a rural versus urban divide, but rather an intra-urban segregation where connectivity quality follows the geography of wealth. Climate events and infrastructure failures disproportionately affect peri-urban zones. For instance, during documented blackouts on 5 March and 27 July 2025, affecting 495,290 users, mobile-dependent neighborhoods lost connectivity for hours whilst fibre-optic zones remained resilient – demonstrating how infrastructure inequality translates directly into differential climate vulnerability. At last, this problem results in education exclusion, employment barriers and policy failure. Our contribution to ITU's UMC goals consist on identifying "second-generation digital deserts" – populations technically counted as "connected" in official statistics but functionally excluded by connectivity so precarious it prevents meaningful participation in digital society. Our methodology provides evidence that universal connectivity requires addressing not just coverage gaps, but intra-urban inequality masked by aggregate statistics.
unizulu team	South Africa	University of Zululand	Shashansk Agarwal	The project addresses the usage gap in Universal and Meaningful Connectivity, focusing on populations that have access to broadband networks but are unable to use the Internet in a meaningful way. These digital deserts emerge not from lack of infrastructure, but from intersecting barriers such as affordability constraints, limited access to appropriate devices, insufficient digital skills, and safety concerns. Addressing this challenge is essential for achieving inclusive digital transformation and meeting ITU's UMC targets for 2030.
URA	Albania	NIV Albania, Lufthansa Industry Solutions , Idealdevs, Hello Albania	Esperanza Magpantay	Our project focuses on the affordability barrier and the exclusion of vulnerable subpopulations from meaningful Internet use across the Western Balkans.In many parts of the region, network coverage is technically present, yet large segments of the population remain offline or under-connected because the cost of access is too high relative to their income or because they lack appropriate devices and digital skills. We want to uncover these less visible barriers, where connectivity exists on paper but is not realistically usable for many people.We will focus on what can be described as hidden digital exclusion. These are communities where infrastructure has reached, but connectivity is still out of reach for low-income households, women, elderly people, and students. By analysing affordability together with demographic and social factors, we aim to show that the problem is not only about building networks, but also about ensuring that people can actually afford and use them in their daily lives.
Vibe_Coders	Bangladesh	Gono Bishwabidyalay	Nathan Menton	Not everyone gets a fair chance online - especially when it comes to using tech or finding ways to practice communication. This gap hits hard in poorer areas, where students, people looking for jobs, and those learning English often lack basic support. Tools that help build these skills? They're missing or too tough to reach. Without them, confidence and opportunity shrink fast. When people can reach the web, some still stay on the edges because they do not feel sure enough to handle job talks, school tasks, or work messages online. Without practice in these areas, being connected means less than it could. That gap - between having access and knowing how to use it - blocks real progress. A single spark can start a fire. This tool runs on artificial intelligence, yet needs little internet to work. Because it works smoothly even on slow connections, anyone can join. Talking becomes easier when mistakes feel safe. Feedback shows up right away, shaped by how you speak. Confidence grows not from perfection but from trying again. Connection matters most when everyone can step in. Meaning appears where inclusion leads.